Gravitation

This chapter is mostly derived from the reference Thornton, Marion - Classical Dynamics of Particles and Systems.

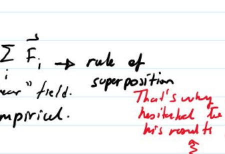
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Isaac Newton formulated and numerically checked the gravitational law in 1687. According to that:

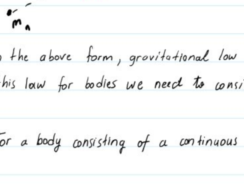
"Each mass particle attracts every other particle in the universe with a force that varies directly as the product of two masses and inversely as the square of the distance between them."

mathematically

$$\vec{F} = -G\frac{mM}{r^2}\hat{r}$$



This law was first experimentally verified in 1768 by Cavendish. G was determined.



$\vec{F}_{net} = \sum_i \vec{F}_i$ → rule of superposition

it is a "linear" field. empirical.

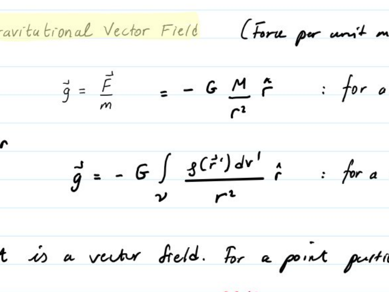
That's why Newton hesitated to publish his results for 20 years.

☆ In the above form, gravitational law strictly applies only to point particles. If we want to extend this law for bodies we need to consider the rule of superposition.

☆ For a body consisting of a continuous distribution of matter, the above sum becomes an integral:

$$\vec{F} = -Gm\int_V \frac{\rho(\vec{r}')dv'}{r^2}\hat{r}$$

$\rho(\vec{r}')$: mass density.
dv': element of volume.
$\vec{r}'$: location of dv' w.r.t. the origin.



$$d\vec{F} = -\frac{Gm[\rho(\vec{r}')dv']}{r^2}$$

Gravitational Vector Field

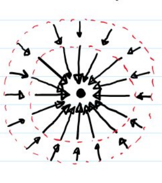
(Force per unit mass in units $\frac{[L]}{[T^2]} \xrightarrow{SI} \frac{m}{s^2}$) ← acceleration.

$$\vec{g} = \frac{\vec{F}}{m} = -G\frac{M}{r^2}\hat{r} \quad : \text{for a point particle of mass } M.$$

or

$$\vec{g} = -G\int_V \frac{\rho(\vec{r}')dv'}{r^2}\hat{r} \quad : \text{for a continuous mass distribution.}$$

It is a vector field. For a point particle M, it looks like:



Near the surface of earth the magnitude of $\vec{g}$ is just the quantity that we call the gravitational acceleration constant. It can be measured with a simple pendulum. $\omega = \sqrt{\frac{g}{\ell}} = 2\pi f = \frac{2\pi}{T} \Rightarrow T = 2\pi\sqrt{\frac{\ell}{g}}$

Gravitational Potential

Gravitational vector field $\vec{g}(\vec{r}) = -\frac{GM}{r^2}\hat{r}$ is conservative.

i.e. $\oint \vec{g}\cdot d\vec{r} = 0$ or $W' = \int_1^2 \vec{g}\cdot d\vec{r}$ only depends on 1 and 2; not the path connecting them. (W': work done per unit mass)

$\Rightarrow$ $\vec{g} = -\vec{\nabla}\phi$ where $\phi(\vec{r})$ is called gravitational potential.

ϕ is in units of $\frac{Fm}{kg}$ or $\frac{J}{kg}$

$\vec{\nabla} \equiv \frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\hat{\phi}$ (the last two terms are 0)

$$\vec{\nabla}\phi = \frac{d\phi}{dr}\hat{r} = G\frac{M}{r^2}\hat{r} \Rightarrow \int d\phi = \int G\frac{M}{r^2}dr + C \Rightarrow \phi(r) = -G\frac{M}{r}$$

(we choose it (C) zero.)

~ THE POTENTIAL OF A POINT MASS ~

☆ The superposition rule applies for the potential too: $\vec{g}_{tot} = \frac{\vec{F}_{tot}}{m} = \frac{\vec{F}_1}{m} + \dots + \frac{\vec{F}_n}{m} = -\vec{\nabla}\phi_1 - \dots - \vec{\nabla}\phi_n = -\vec{\nabla}(\phi_1 + \phi_2 + \dots + \phi_n)$

Therefore, for continuous mass distributions:

$$\phi(r) = -G\int_V \frac{\rho(\vec{r}')}{r}dv' \quad \text{Potential of a volume mass distribution}$$

$$\phi(r) = -G\int_S \frac{\sigma(\vec{r}')}{r}da' \quad \text{" surface " "}$$

$$\phi(r) = -G\int_P \frac{\lambda(\vec{r}')}{r}ds' \quad \text{" line " "}$$

☆ The work per unit mass must be done by an agent on a body in a gravitational field to displace a body a distance $d\vec{r}$.

$\Rightarrow$ The force (per unit mass) applied by the agent must have the same magnitude as the gravitational force but must be opposite to it. $\vec{F}'_{ag} = -\vec{g}$ (force per unit mass)

reminder (see mathematical tools chapter)

$\Rightarrow dW' = -\vec{g}\cdot d\vec{r} = \vec{\nabla}\phi\cdot d\vec{r} = d\phi = \sum_i \frac{\partial\phi}{\partial x_i}dx_i$

$\Rightarrow$ The work per unit mass done by the agent is equal to the difference in potential at two points.

$$dW' = d\phi = \phi_2 - \phi_1 = GM\left(-\frac{1}{r_2} + \frac{1}{r_1}\right)$$

$\Rightarrow$ If $r_2 > r_1$, $dW' > 0$; if the final position is farther from the source of mass agent does a work on a unit mass.

An interpretation:

$\Rightarrow$ An agent does work on a unit mass as it takes the mass from a finite distance to the source to infinity, the work done is "stored" as the potential energy in the field.

★ The potential is zero at infinity. Then, we may interpret ϕ at any point r to be the work per unit mass required to bring a particle from infinity to point r.

$$dW' = GM\left(-\frac{1}{r_2} + \frac{1}{r_1}\right) = -\frac{GM}{r} \quad (r_2 \to r,\ r_1 \to \infty)$$

$\Rightarrow$ The work done by an external agent on a unit mass to bring it from ∞ to r is negative; "the force takes the field's energy away".

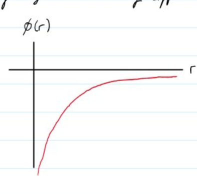
☆ The potential energy is equal to the mass of the body multiplied by the potential ϕ. If U is the potential energy, then

$$U = m\phi$$

and the force on a body is given by the negative of the gradient of the potential energy of that body

$$\vec{F} = -\vec{\nabla}U$$

☆ The potential and the potential energy is increased when work is done on the body. The potential according to our definition is always negative and only approaches its maximum value (zero), as r tends to infinity.

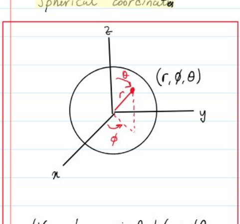


☆ Whenever a particle is created in a gravitational field, a certain potential energy exists. One way of interpreting this energy is that by considering it to be stored in the "field". However, if we are examining a single particle and its potential, it is customary to speak of the potential energy of the body. Sometimes we may consider the source mass itself to have an intrinsic potential energy. This potential energy is equal to the gravitational energy released when the body was formed or, conversely, is equal to the energy that must be supplied to disintegrate the mass over the sphere at infinity. For example, when interstellar gas condenses to form a star, the gravitational energy released goes largely into the initial heating of the star. As the temperature increases, energy is radiated away as electromagnetic radiation.

$\Rightarrow$ In all problems we treat in here, the structure of the bodies is considered to remain unchanged during the process we are studying. Thus, there will not be any change in the intrinsic potential energy.

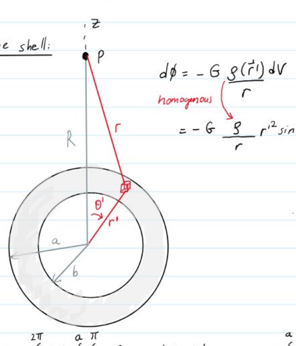
Example: What is the gravitational potential both inside and outside a spherical homogenous shell of inner radius b and outer radius a?

Spherical coordinates



$dV = dr\ r\sin\theta\, d\phi\ r\, d\theta = r^2\sin\theta\, dr\, d\phi\, d\theta$

Outside the shell:



$d\phi = -G\frac{\rho(\vec{r}')dV}{r}$ (homogenous) $= -G\frac{\rho}{r}r'^2\sin\theta' dr' d\phi' d\theta'$

$$\phi(R) = -G\rho\int_0^{\pi}\int_0^{2\pi}\int_b^a \frac{r'^2}{r}\sin\theta' dr' d\theta' d\phi' = -G\rho\int_0^{2\pi}d\phi'\int_b^a\int_0^{\pi}\frac{r'^2}{r}\sin\theta' dr' d\theta' = -G\rho 2\pi\int_b^a\int_0^{\pi}\frac{r'^2}{r}\sin\theta' dr' d\theta'$$

- R constant, and R, r', r, θ' are connected. According to law of cosines:

$$R^2 + r'^2 - 2Rr'\cos\theta' = r^2$$

- If we want to describe the change in r in terms of the changes in θ', we can consider the r derivative of above equation for a given r' (i.e., if we keep r' constant, to change r, we need to change θ'):

$$\frac{d}{dr}\left(R^2 + r'^2 - 2Rr'\cos\theta'\right) = \frac{d}{dr}(r^2)$$

$$-2Rr'\sin\theta'\frac{d\theta'}{dr} = 2r \Rightarrow \frac{\sin\theta'}{r}d\theta' = \frac{dr}{r'R} \quad \text{substitude this in the integral.}$$

$$\phi(R) = -G\rho 2\pi\int_b^a\int_{r_{min}}^{r_{max}}\frac{r'^2}{r'R}dr\,dr' = -G\rho\frac{2\pi}{R}\int_b^a r'dr'\int_{r_{min}}^{r_{max}}dr$$

☆ The boundaries of $\int_{r_{min}}^{r_{max}}dr$ integral depend on the location of point P. If P is outside the shell

$$\phi(R>a) = -G\rho\frac{2\pi}{R}\int_b^a r'dr'\int_{R-r'}^{R+r'}dr = -G\rho\frac{2\pi}{R}\int_b^a r'(R + r' - R + r')dr' = -G\rho\frac{2\pi}{R}\int_b^a 2r'^2dr'$$

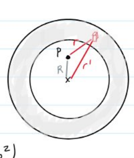
$$= -G\rho\frac{4\pi}{R}\left(\frac{b^3}{3} - \frac{a^3}{3}\right) = -\frac{4}{3}\frac{\pi G\rho}{R}(a^3 - b^3)$$

The total mass of the shell:

$$M = \frac{4}{3}\pi\rho a^3 - \frac{4}{3}\pi\rho b^3 = \frac{4}{3}\pi\rho(a^3 - b^3) \Rightarrow \phi(R>a) = -\frac{GM}{R}$$

→ scales with $\sim \frac{1}{R}$

- If point P lies inside the shell, then



$$\phi(R<b) = -\frac{G\rho 2\pi}{R}\int_b^a r'dr'\int_{r'-R}^{r'+R}dr = -\frac{G\rho 2\pi}{R}\int_b^a r'(r' + R - r' + R)dr'$$

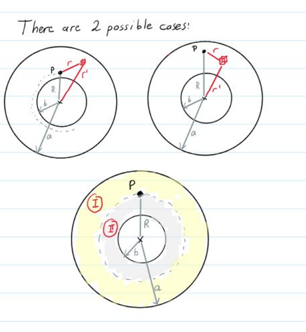
$$= -\frac{G\rho 2\pi 2R}{R}\int_b^a r'dr' = -G\rho 4\pi\left(\frac{a^2}{2} - \frac{b^2}{2}\right) = -2\pi\rho G(a^2 - b^2)$$

$$\phi(R<b) = -2\pi\rho G(a^2 - b^2) \quad \rightarrow \text{constant}$$

- Potential within the shell: $\phi(b<R<a)$

→ We can divide the mass distribution into 2 parts: (I) and (II)

There are 2 possible cases:



→ To calculate the potential due to part (I) can be directly obtained by using the result for $\phi(R<b)$ but changing the lower limit of the final integral by R:

$$\phi_I = -G\rho 4\pi\int_R^a r'dr' = -G\rho 4\pi\left(\frac{a^2}{2} - \frac{R^2}{2}\right)$$

→ The potential at point P due to part (II) can be obtained by considering $\phi(R>a)$ but replacing the upper limit in the final integral by R:

$$\phi_{II} = -G\rho\frac{2\pi}{R}\int_b^R 2r'^2dr' = -G\rho\frac{4\pi}{R}\left(\frac{R^3}{3} - \frac{b^3}{3}\right)$$

$$\phi(b<R<a) = \phi_I + \phi_{II} = -G\rho 4\pi\left(\frac{a^2}{2} - \frac{R^2}{2} + \frac{R^2}{3} - \frac{b^3}{3R}\right)$$

$$\phi(b<R<a) = -4\pi\rho G\left(\frac{a^2}{2} - \frac{b^3}{3R} - \frac{R^2}{6}\right)$$

$$\phi(R=b) = -4\pi\rho G\left(\frac{a^2}{2} - \frac{b^2}{3} - \frac{b^2}{6}\right)$$

☆ To calculate the gravitational vector field, we use $\vec{g} = -\vec{\nabla}\phi = -\frac{d\phi}{dR}\hat{r}$

$$\vec{g}(b<R<a) = -\frac{d}{dR}\left[-4\pi\rho G\left(\frac{a^2}{2} - \frac{b^3}{3R} - \frac{R^2}{6}\right)\right] = 4\pi\rho G\left(\frac{b^3}{3R^2} - \frac{R}{3}\right)\hat{r} = -4\pi\rho G\left(\frac{R}{3} - \frac{b^3}{3R^2}\right)\hat{r}$$

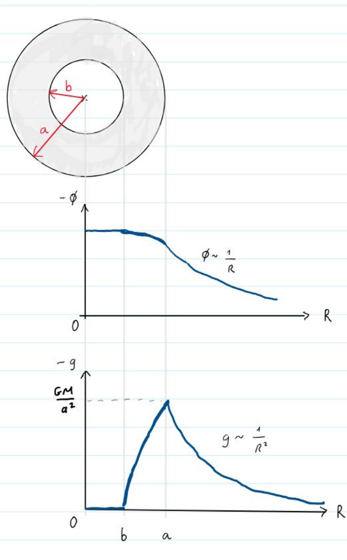
$$\vec{g}(R<b) = 0$$

$$\vec{g}(R>a) = -\frac{d}{dR}\left(-\frac{GM}{R}\right)\hat{r} = -\frac{GM}{R^2}\hat{r}$$

$$-4\pi\rho G\left(\frac{a}{3} - \frac{b^3}{3a^2}\right)$$

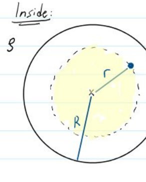
$$-\frac{4\pi\rho G}{3}\left(\frac{a^3}{a^2} - \frac{b^3}{a^2}\right)$$

$$-\underbrace{\frac{4\pi}{3}\rho(a^3 - b^3)}_{M}\frac{G}{a^2}$$



Ex: Find the gravitational field inside and outside a spherical homogenous mass distribution.

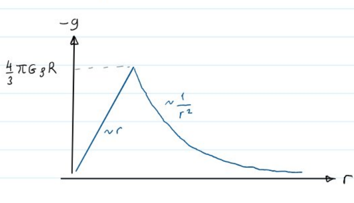
Inside:



$$\vec{g} = -\frac{G}{r^2}\underbrace{\left(\frac{4}{3}\pi r^3\right)\rho}_{\text{total mass in yellow area}}\hat{r} = -\frac{4}{3}\pi G\rho r\hat{r} \quad : \text{increases linearly.}$$

Outside

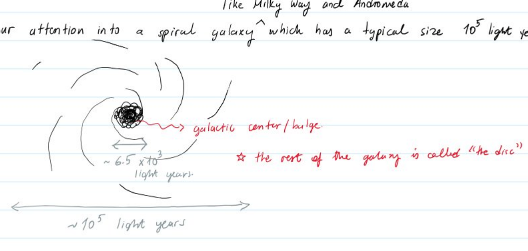
$$\vec{g} = -\frac{G}{r^2}\underbrace{\left(\frac{4}{3}\pi R^3\right)\rho}_{\text{total mass of the sphere}}\hat{r} = -\frac{4}{3}\pi G\rho R^3\frac{1}{r^2} \quad : \text{decreases quadratically.}$$



Ex: How are the orbital speeds of the planets scaled with the distance to the sun by considering the fact that the mass of the sun M accounts for 99.8 % of our solar system's mass

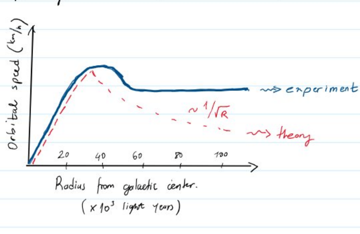
$$\frac{GMm}{R^2} = \frac{mv^2}{R} \Rightarrow v^2 = \frac{GM}{R} \Rightarrow v = \sqrt{\frac{GM}{R}} \sim \frac{1}{\sqrt{R}} \quad : \text{We also confirm this with observations.}$$

Now let us turn our attention into a spiral galaxy (like Milky Way and Andromeda) which has a typical size 10^5 light years.



☆ the rest of the galaxy is called "the disc"

If we want to describe the orbital speeds of masses around the galactic bulge, we can consider the equation we derived for the solar system since most of the mass of the galaxy is enclosed in the bulge (that's what astronomers observe) However, if these orbital speeds are observed the following behavior is found (blue curve)



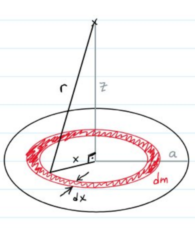
We would expect the orbital speed to decrease as $1/\sqrt{R}$. However, v is a constant function of distance from the center. This can only happen if mass M of the galaxy itself is a linear function of R $M(R) \sim R$ in the above equation:

$$\frac{GM(R)m}{R^2} = \frac{mv^2}{R} \Rightarrow G\alpha = v^2 \Rightarrow v = \sqrt{\alpha G} : \text{constant.}$$

($M(R) \sim R$; α: scaling factor for M.)

Astrophysicist conclude from this result that for many galaxies there must be matter other than that observed, and this "unobserved matter" often called "dark matter". must account for more than 90% of the known mass in the universe.

Ex: Consider a thin uniform disc of mass M and radius a. Find the force on a mass m located along the axis of the disc.



$$d\phi = -G\frac{dm}{r} = -G\frac{\sigma da}{r} = -G\sigma\frac{2\pi x\, dx}{r}$$

$$= -\frac{G\sigma 2\pi x\, dx}{\sqrt{z^2 + x^2}}$$

$$\phi(z) = -G\sigma\pi\int_0^a\frac{2x\,dx}{\sqrt{z^2 + x^2}} \qquad u \equiv z^2 + x^2 \to du = 2x\,dx$$

$$= -G\sigma\pi\int_{z^2}^{z^2+a^2}u^{-1/2}du = -G\sigma\pi\left.\frac{u^{1/2}}{\frac{1}{2}}\right|_{z^2}^{z^2+a^2}$$

$$= -2G\sigma\pi\left(\sqrt{z^2 + a^2} - z\right)$$

☆ From symmetry, we know that $\vec{g}$ will have only z component.

$$g_z = -\frac{d}{dz}\left[-2G\sigma\pi\left(\sqrt{z^2 + a^2} - z\right)\right]$$

$$= 2\pi G\sigma\left[\frac{1}{2}(z^2 + a^2)^{-1/2}2z - 1\right]$$

$$= 2\pi G\sigma\left[\frac{z}{\sqrt{z^2 + a^2}} - 1\right]$$

$$\Rightarrow F_z = 2\pi mG\sigma\left[\frac{z}{\sqrt{z^2 + a^2}} - 1\right]$$